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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
To investigate if junctions tend to happen in non-stem regions, the number of junctions occurring in base-paired positions were calculated and compared with a background distribution for the numbers of junctions located in stems derived from a 1000-time random sampling of the same number of sites along the viral RNA strand.
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.
Cell Line Authentication

not detected.

Data Information

Availability: Identification of insertions in SARS-CoV-2 genomes The SARS-CoV-2 genomes available at GISAID (https:www.gisaid.org) on 2021-12-17 were downloaded for analysis (n=6,163,073).

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental Models: Cell Lines			
Analysis of the expression level of host genes observed in chimeric reads Gene expression profiles for two SARS-CoV-2 infected Caco-2 cell line samples (GSM4477888, GSM4477889), two SARS-CoV-2 infected Calu-3 cell line samples (GSM4477962, GSM4477963), and three SARS-CoV-2 infected Vero-6 cell line samples (GSM4916368, GSM4916369, GSM4916370) were downloaded from the GEO database.	Caco-2		Suggestion: RRID:CVCL_0025(link)
	Calu-3		Suggestion: RRID:CVCL_0609(link)
	Vero-6		
Software and Algorithms			
METHODS Identification of host-virus chimeric reads in SARS-CoV-2 direct-RNA seq data The nanopore direct RNA-seq data from SARS-CoV-2 infected cell lines were downloaded from the NCBI SRA database (Supplementary Table 1).	NCBI SRA		
All reads were quality trimmed using NanoFilt v2.8.0 (41), to remove the first 50 nucleotides of each read and require an average quality score of at least 10 over the length of the read.	NanoFilt		Suggestion: NanoFilt (RRID:SCR_016966)(link)
The short reads were first preprocessed with fastp v0.23.1 (44) and then mapped to the chimeric reads from the same samples by using Minimap2 v2.23 (42) with options -ax sr -w 5 to tolerate the high error rate of the Nanopore direct RNA-sequencing reads (45).	Minimap2		Suggestion: Minimap2 (RRID:SCR_018550)(link)
They were searched in the NCBI non-redundant nucleotide database and a collection of coronavirus genomes with BLASTN (E-value<=1e-2) (32) to explore their possible origins.	BLASTN		
The raw sequencing reads were processed for quality control using fastp v0.23.1 (44) with default parameters and mapped to the SARS-CoV-2 reference genome using BWA mem v0.7.17 (52).	BWA		Suggestion: BWA (RRID:SCR_010910)(link)
The alignment files were converted to FASTQ format using samtools fastq command v1.14 (54) and re-aligned to the consensus genomes with or	samtools		
	bowtie		

without the insertion using bowtie v2.4.4 (45) (parameter: --xeq).			

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
To further examine which types of structural elements are over- or under-presented at junction sites in virus-host chimeric reads and in host-virus chimeric reads, one-sided Fishers exact test was performed.	Fishers exact test
To evaluate whether the expression level of the host protein-coding genes in chimeric reads is significantly greater than the expression level of the background gene set, a one-sided Mann-Whitney U test was performed for each sample.	Mann-Whitney U test

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq